



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

**MCA
I / II Year
Dec.2024/Jan.2025
2022 Scheme**

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CBCS SCHEME

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MMC101

First Semester MCA Degree Examination, Dec.2024/Jan.2025 Programming and Problem Solving in C

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L:Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain the characteristics of C programming language		8	L2	CO1
	b.	Explain the structure of C program		5	L2	CO1
	c.	List the uses of C Language.		7	L2	CO1
OR						
Q.2	a.	Explain the various forms of if statement with examples		12	L2	CO1
	b.	With example explain while,do-while,for loops		8	L2	CO1
Module – 2						
Q.3	a.	Define array. Give syntax of 1-D array.Explain the operations on array.		10	L2	CO2
	b.	Give 2-D array syntax. Write C program to multiply two matrices.		10	L2	CO3
OR						
Q.4	a.	Define string with example. Explain the string taxonomy		08	L2	CO3
	b.	Explain the various string operations		05	L2	CO3
	c.	Write a C program search element in an array using linear search.		07	L2	CO3
Module – 3						
Q.5	a.	Define a Function. Differentiate between call by value and call by reference.		8	L3	CO4
	b.	Define recursive function .write a C program to find the factorial of function using recursive function.		7	L3	CO4
	c.	Write a C program to swap a two numbers by call by value.		5	L3	CO4
OR						
Q.6	a.	Write a C program to find the mean of N numbers using arrays and pointers		10	L3	CO3
	b.	Write a C program to add two matrices using pointers.		10	L3	CO3
Module – 4						
Q.7	a.	Define Structure. Give syntax of Structure.		5	L2	CO5
	b.	Write a C program using structure to read and display student information.		10	L2	CO5
	c.	Explain nested structure with example.		5	L2	CO5
OR						
Q.8	a.	Define Union. Give the syntax of Union.		5	L2	CO5
	b.	Explain the various storage classes		10	L2	CO5
	c.	Write a short note typedef.		5	L2	CO5

Module – 5					
Q.9	a.	Define a File. List and explain the operations on file	10	L2	CO1
	b.	Write a note on	10	L2	CO1
1. fscanf() 3.fgetc() 2. fgets() 4. fread()					
OR					
Q.10	a.	Write a note on :	10	L2	CO1
		1. Fprintf() 2. Fputs() 3. Fputc() 4. Fwrite()			
	b	Write a note on :	10	L2	CO1
		1. Fseek() 2. Ftell() 3. Fgetpos() 4. Fsetpos()			

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First Semester MCA Degree Examination, Dec.2024/Jan.2025

Discrete Mathematics and Graph Theory

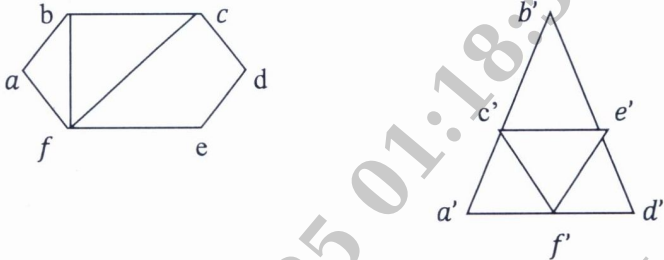
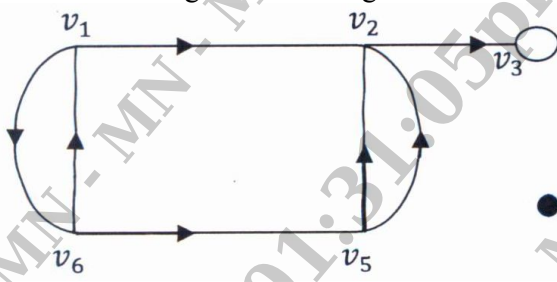
Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 7\}$, $B = \{4, 5, 6, 7\}$ and $C = \{1, 3, 6\}$. Compute the following: (i) $\overline{A \cup C}$ (ii) $\overline{A} \cap \overline{B}$ (iii) $A \cap B \cap C$ (iv) $B - A$ (v) $A - B$	6	L1	CO1
	b.	Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{6, 7, 8, 9, 10\}$ and $f: A \rightarrow B$ be a function defined by $f = \{(1,7)(2,7)(3,8)(4,6)(5,9)(6,9)\}$. Determine $f^{-1}(6)$ and $f^{-1}(9)$. Also if $B_1 = \{7, 8\}$, $B_2 = \{8, 9, 10\}$ then find $f^{-1}(B_1)$ and $f^{-1}(B_2)$.	7	L2	CO1
	c.	Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$.	7	L2	CO1
OR					
Q.2	a.	For any two sets A and B , prove the Demorgan's laws.	6	L1	CO1
	b.	State pigeon-hole principle. Show that if 50 books in a library contain a total of 27551 pages, one of the books must have atleast 552 pages.	7	L2	CO1
	c.	In a class of 52 students, 30 are studying C++, 28 are studying pascal and 13 are studying both languages. How many in this class are studying at least one of these languages? How many are studying neither of these languages?	7	L2	CO1
Module – 2					
Q.3	a.	Define tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology	7	L2	CO2
	b.	Write the converse, inverse and the contra positive of the conditional statement: "If oxygen is a gas then Gold is compound".	6	L2	CO2
	c.	Prove the following is valid argument : $\begin{array}{l} p \rightarrow r \\ \sim p \rightarrow q \\ q \rightarrow s \\ \hline \therefore \sim r \rightarrow s \end{array}$	7	L2	CO2
OR					
Q.4	a.	Prove the following using the laws of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$	7	L2	CO2
	b.	Negate and simplify: (i) $\forall x, [p(x) \wedge \sim q(x)]$. $\exists x, [\{p(x) \vee q(x)\} \rightarrow r(x)].$	6	L2	CO2
	c.	Give the direct proof of the following statement "If n is an odd integer, then n^2 is odd."	7	L2	CO2
Module – 3					
Q.5	a.	Define graph and explain the types of graph.	8	L1	CO3

	b.	Prove that the number of vertices of odd degree in a graph is always even.	6	L2	CO3
	c.	Define isomorphic graph and verify the following graphs are isomorphic or not. 	6	L2	CO3
OR					
Q.6	a.	Explain the following graphs: (i) Bi- partite graph (ii) Sub graphs (iii) Walk (iv) Path	10	L1	CO3
	b.	Prove that a simple graph with n vertices and K components can have at most $(n-k)(n-k+1)/2$ edges.	10	L2	CO3
Module – 4					
Q.7	a.	State and prove necessary condition of a graph to be a Euler graph.	10	L2	CO4
	b.	List and explain the different operations on graph.	10	L2	CO4
OR					
Q.8	a.	Define digraph. Find the indegree and outdegree of the following graph: 	8	L2	CO4
	b.	Illustrate the travelling salesman problem using a graph.	6	L2	CO4
	c.	List and explain different digraphs and binary relations.	6	L2	CO4
Module – 5					
Q.9	a.	Prove that every tree with two or more vertices is 2- Chromatic	10	L2	CO5
	b.	Explain the following for chromatic polynomial: (i) Finding a maximal independent set. (ii) Finding all maximal independent set.	10	L2	CO5
OR					
Q.10	a.	Prove that the vertices of every planar graph can be properly colored with five colors.	10	L2	CO5
	b.	Explain the Greedy coloring algorithm.	10	L2	CO5

First Semester MCA Degree Examination, Dec.2024/Jan.2025
Database Management Systems (DBMS)

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Define the following terms: i. DBMS ii. Entity & Attribute iii. Relational data model iv. Schema and Schema Diagram v. Primary key and Foreign key	10	L1	CO1	
	b.	Discuss the different applications of DBMS.	5	L2	CO1	
	c.	Explain three schema architecture with neat diagram.	5	L2	CO1	
OR						
Q.2	a.	Explain components of DBMS.	10	L2	CO1	
	b.	Discuss the different types of Relationship types.	5	L2	CO1	
	c.	Draw ER-Diagram for Company database which contains entity type Employee, Department, Project and Dependent.	5	L3	CO1	
Module – 2						
Q.3	a.	Explain the following relational algebra operations i. Selection ii. Projection	10	L2	CO2	
	b.	Describe the following DDL and DML commands. i. Create ii. Insert iii. Delete iv. Update v. Drop	10	L2	CO2	
OR						
Q.4	a.	Explain the following clauses i. select ...From...Where clause ii. Group by and Having clause	10	L2	CO3	
	b.	Elaborate the importance of views.	5	L2	CO3	
	c.	Discuss about Procedures.	5	L2	CO3	
Module – 3						
Q.5	a.	Explain 1NF and 2NF with an example.	10	L2	CO3	
	b.	Discuss 3NF and Boyce codd with an example.	10	L2	CO3	
OR						
Q.6	a.	Explain 4NF and 5NF with an example.	10	L2	CO3	
	b.	Discuss the following with an Example. i. Functional dependency ii. Dependency Preservation Property	10	L2	CO3	
Module – 4						
Q.7	a.	Describe the following i. ACID Properties of transaction ii. Different states for transaction execution	10	L2	CO2	
	b.	Discuss two-phase locking system with an example.	10	L2	CO2	
OR						
Q.8	a.	Explain the implementation of Isolation level.	10	L2	CO2	
	b.	Discuss Multiple Granularity with an example.	10	L2	CO2	
Module – 5						
Q.9	a.	How the log can be used to recover from a system crash and to roll back transactions during normal operation?	10	L1	CO2	
	b.	Illustrate Checkpoints and Fuzzy Check pointing with an example.	10	L3	CO2	
OR						

1 of 2

Q.10	a.	Describe the Buffer Management with an example.	10	L2	CO2
	b.	Define Undo and Redo options. The log states are mentioned below. Determine the use of Undo and Redo options to ensure the atomicity in below mentioned examples. <T₀ start> < T₀ , A, 1000, 950> < T₀ , B, 2000, 2050> (i) <T₀ start> < T₀ , A, 1000, 950> < T₀ , B, 2000, 2050> <T₀ commit> <T₁ start> < T₁ , C, 700, 600> (ii) <T₀ start> < T₀ , A, 1000, 950> < T₀ , B, 2000, 2050> <T₀ commit> <T₁ start> < T₁ , C, 700, 600> <T₁ commit> (iii)	10	L3	CO2

First Semester MCA Degree Examination, Dec.2024/Jan.2025 Operating System

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C																								
Q.1	a.	What is the significance of Operating System? Illustrate various services provided by the Operating System.	10	L2	CO1																								
	b.	What is the purpose of system calls? Describe different types of system calls used in Operating system with examples.	10	L2	CO1																								
OR																													
Q.2	a.	Illustrate the following operating system architectures with a neat diagram: (i) Microkernel (ii) Layered	10	L2	CO1																								
	b.	Illustrate with a neat diagram various states of process. Also discuss the significance of process control block (PCB).	10	L2	CO1																								
Module – 2																													
Q.3	a.	“CPU scheduling ensures proper execution of processes”. Justify. Illustrate different scheduling criteria used by CPU scheduling algorithms.	10	L2	CO1																								
	b.	Discuss how dining philosophers problem is solved using semaphores.	10	L3	CO1																								
OR																													
Q.4	a.	What do you mean by Critical Section Problem? Explain the solution to the critical-section problem using mutex locks.	10	L2	CO1																								
	b.	Consider the set of processes with Arrival Time, CPU burst time (in milliseconds) and priority as shown below. (Lower number represents higher priority). <table border="1"><thead><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr></thead><tbody><tr><td>P1</td><td>0</td><td>10</td><td>3</td></tr><tr><td>P2</td><td>1</td><td>1</td><td>1</td></tr><tr><td>P3</td><td>2</td><td>2</td><td>4</td></tr><tr><td>P4</td><td>3</td><td>1</td><td>5</td></tr><tr><td>P5</td><td>4</td><td>5</td><td>2</td></tr></tbody></table> Draw the Gantt Chart and calculate the Average waiting time and Average Turnaround time using 1) SJF Scheduling (Non Pre-emptive) 2) Priority Scheduling (Non Pre-emptive) (Note: Consider Arrival Time for both algorithms.)	Process	Arrival Time	Burst Time	Priority	P1	0	10	3	P2	1	1	1	P3	2	2	4	P4	3	1	5	P5	4	5	2	10	L3	CO1
Process	Arrival Time	Burst Time	Priority																										
P1	0	10	3																										
P2	1	1	1																										
P3	2	2	4																										
P4	3	1	5																										
P5	4	5	2																										

1 of 2

Module – 3

Q.5	a.	Illustrate deadlocks with their necessary conditions.	10	L2	CO2																																																																															
	b.	Describe the working principles of Banker's algorithm for the following snapshot and find either the system is in safe state or not. <table><tr><td></td><td colspan="4">Allocation</td><td colspan="4">Max</td><td colspan="4">Available</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td><td>A</td><td>B</td><td>C</td><td>D</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>P₀</td><td>0</td><td>0</td><td>1</td><td>2</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>0</td></tr><tr><td>P₁</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>7</td><td>5</td><td>0</td><td colspan="4" rowspan="4"></td></tr><tr><td>P₂</td><td>1</td><td>3</td><td>5</td><td>4</td><td>2</td><td>3</td><td>5</td><td>6</td></tr><tr><td>P₃</td><td>0</td><td>6</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>1</td><td>4</td><td>0</td><td>6</td><td>5</td><td>6</td></tr></table>		Allocation				Max				Available					A	B	C	D	A	B	C	D	A	B	C	D	P ₀	0	0	1	2	0	0	1	2	1	5	2	0	P ₁	1	0	0	0	1	7	5	0					P ₂	1	3	5	4	2	3	5	6	P ₃	0	6	3	2	0	6	5	2	P ₄	0	0	1	4	0	6	5	6	10	L2	CO2
	Allocation				Max				Available																																																																											
	A	B	C	D	A	B	C	D	A	B	C	D																																																																								
P ₀	0	0	1	2	0	0	1	2	1	5	2	0																																																																								
P ₁	1	0	0	0	1	7	5	0																																																																												
P ₂	1	3	5	4	2	3	5	6																																																																												
P ₃	0	6	3	2	0	6	5	2																																																																												
P ₄	0	0	1	4	0	6	5	6																																																																												

OR

Q.6	a.	Discuss deadlock detection with a neat diagram.	10	L2	CO2
	b.	Explain different methods used for recovering from a deadlock in an operating system.	10	L2	CO2

Module – 4

Q.7	a.	Describe in detail the concept of Paging with a neat diagram.	10	L3	CO3
	b.	Differentiate between internal and external fragmentation.	10	L2	CO3

OR

Q.8	a.	Consider the page reference string: 1,0,7,1,0,2,1,2,3,0,3,2,4,0,3,6,2,1 for a memory with three frames. Determine the number of page faults using the FIFO, Optimal, and LRU replacement algorithms. Which algorithm is most efficient?	10	L3	CO3
	b.	Interpret the concepts of demand paging with neat diagram.	10	L2	CO3

Module – 5

Q.9	a.	Illustrate the following access methods. i) Sequential access ii) Direct access	08	L2	CO3
	b.	Illustrate in detail the various operations performed on a file.	08	L2	CO3
	c.	Explain the following: i) Bit vector ii) Linked list	04	L2	CO3

OR

Q.10	a.	Illustrate various levels of directory structures.	10	L2	CO3
	b.	List the different file allocation methods and explain any two methods in detail.	10	L2	CO3

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MMC105

First Semester MCA Degree Examination, Dec.2024/Jan.2025 Web Technologies

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define HTTP. Explain the different Phases of HTTP.	10	L2	CO1
	b.	Discuss the basic structure of XHTML document. Also explain the rules to be followed to make use of HTML elements in XHTML document.	10	L2	CO1
OR					
Q.2	a.	Briefly explain the following: 1. URL 2. MIME 3. web server 4. web browser	10	L2	CO1
	b.	Explain the following tags with examples 1. Heading tag 2. Hypertext link tag 3. Image tag 4. Progress tag	10	L2	CO1
Module – 2					
Q.3	a.	Discuss on the different ways of including CSS style information to a HTML document.	10	L2	CO2
	b.	Name any five CSS selectors and explain their uses with a suitable example.	10	L3	CO2
OR					
Q.4	a.	Explain the various ways of creating arrays in javascript. Mention any 5 array methods and explain their use.	10	L2	CO2
	b.	Write javascript program that accepts. 1. Input : A number n Output : The first n Fibonacci numbers 2. Input : A number n Output : A table of numbers from 1 to n and their squares	10	L3	CO2
Module – 3					
Q.5	a.	Explain Document object model (DOM) with an example.	10	L2	CO3
	b.	Write a javascript program to show handling of events from textbox and password elements.	10	L3	CO3
OR					
Q.6	a.	Briefly describe Window object's properties and methods.	10	L2	CO3
	b.	Discuss Event handling. Explain it with an example.	10	L2	CO3
1 of 2					

Module – 4

Q.7	a.	Briefly explain the following with examples: 1. AngularJS Numbers 2. AngularJS Strings 3. AngularJS Objects 4. AngularJS Arrays	10	L2	CO4
	b.	Discuss the use of filters in Angular JS with an example.	10	L2	CO4

OR

Q.8	a.	What is Angular JS? Explain the following Angular JS directives: (i) ng_app (ii) ng_model (iii) ng_bind	10	L2	CO4
	b.	Explain AngularJS expressions. Write an Angular JS program to use expressions.	10	L3	CO4

Module – 5

Q.9	a.	What is Angular JS Services? Explain them with examples.	10	L2	CO4
	b.	Write an Angular JS program to demonstrate client-side form validation.	10	L3	CO4

OR

Q.10	a.	Briefly explain about AngularJS Events with an example.	10	L3	CO4
	b.	Explain AngularJS Forms and its elements.	10	L3	CO4

CBCS SCHEME

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22MCA21

Second Semester MCA Degree Examination, Dec.2024/Jan.2025

Database Management System

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain the characteristics of database approach.		10	L2	CO1
	b.	Briefly discuss the actors on the scene and workers behind the scene.		06	L2	CO1
	c.	What are the advantages of using DBMS approach?		04	L2	CO1
OR						
Q.2	a.	Describe the categories of data models.		06	L2	CO1
	b.	With a neat diagram, explain the three schema architecture.		10	L2	CO1
	c.	Define data independence. Discuss the types of data independence.		04	L2	CO1
Module – 2						
Q.3	a.	Illustrate the relational model constraints and relational database schema with an example.		10	L2	CO4
	b.	Demonstrate the unary relational operations with an example.		06	L3	CO4
	c.	Discuss the relational algebra operations from set theory.		04	L2	CO4
OR						
Q.4	a.	Describe the importance of binary relational operations and additional operational of relational operations.		10	L2	CO4
	b.	Demonstrate the seven steps of relational database design using ER to relational mapping.		10	L3	CO4
Module – 3						
Q.5	a.	What is SQL? Discuss the attribute datatype and domains in SQL.		06	L2	CO2
	b.	Illustrate the features of creating assertion and trigger. Give example.		08	L3	CO2
	c.	Describe the concept of views (virtual tables) in SQL. Also give examples.		06	L3	CO2
OR						
Q.6	a.	Explain the architecture of JDBC main components and classify the drivers in JDBC.		10	L2	CO3
	b.	Discuss the importance of stored procedures and its creation and calling of JDBC and SQLJ.		10	L2	CO3

Module – 4

Q.7	a.	Describe the design guidelines for relation schema.	08	L2	CO3
	b.	Illustrate the process of normalization of relations with 1NF, 2NF, 3NF and BCNF with neat examples.	12	L3	CO5

OR

Q.8	a.	Armstrong's interference rules IR1 through IR2 are sound and complete. Justify this statement.	08	L2	CO5
	b.	Explain the importance of lossless (non additive join property) of a decomposition?	06	L2	CO5
	c.	Discuss the problems with Null values and dangling tuples.	06	L2	CO5

Module – 5

Q.9	a.	Justify concurrency control is necessary. Discuss the problems may encounter.	10	L2	CO5
	b.	Characterize the types of schedules based on serializability.	10	L2	CO5

OR

Q.10	a.	Describe the two-phase locking techniques for concurrency control.	10	L2	CO5
	b.	Demonstrate the granularity of data items and multiple granularity locking.	10	L2	CO5

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22MCA22

Second Semester MCA Degree Examination, Dec.2024/Jan.2025

Object Oriented Programming using Java

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain object oriented principles of Java.	06	L2	CO1
	b.	List any four differences between POP and OOP.	04	L2	CO1
	c.	List and explain advantages and disadvantages of inheritance.	10	L2	CO1
OR					
Q.2	a.	Write a Java program to demonstrate constructor overloading and method overloading.	10	L3	CO2
	b.	Explain Java Access Modifiers with a program.	10	L2	CO2
Module – 2					
Q.3	a.	What is Hierarchical inheritance? Explain with example.	10	L2	CO4
	b.	Write a Java program to perform multilevel inheritance.	10	L3	CO4
OR					
Q.4	a.	Explain early binding and late binding with example.	10	L2	CO3
	b.	Explain abstract class and abstract methods in Java.	10	L2	CO3
Module – 3					
Q.5	a.	List and explain the difference between Abstract class and Interface.	10	L2	CO3
	b.	Briefly explain how to handle multiple inheritance in Java with program.	10	L2	CO3
OR					
Q.6	a.	What are packages? List the advantages of packages in Java.	10	L2	CO1
	b.	i) Explain classpath in Java with proper steps. ii) Difference between path and class path.	10	L2	CO1
Module – 4					
Q.7	a.	Define exception. Explain exception handling mechanism with example.	10	L2	CO5
	b.	Write a Java program to handle multiple Catch Blocks (Exceptions).	10	L3	CO5
OR					
Q.8	a.	List the difference between checked and unchecked exception.	10	L2	CO5
	b.	Explain : i) try ii) catch iii) finally iv) throws	10	L2	CO5
Module – 5					
Q.9	a.	List the difference between AWT and Swing.	10	L2	CO6
	b.	Explain any 5 components of Java Swing.	10	L2	CO6
OR					
Q.10	a.	Write a Java applet program which handles keyboard event.	10	L2	CO6
	b.	With neat diagram explain Java applet life cycle. Also give an example code.	10	L3	CO6

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22MCA24

Second Semester MCA Degree Examination, Dec.2024/Jan.2025 Web Technologies

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain HTTP protocol in detail.	10	L2	CO4
	b.	Explain the following : (i) URL (ii) Browser (iii) MIME (iv) Web Server (v) Internet/web.	10	L2	CO4
OR					
Q.2	a.	With appropriate XHTML tags, create a registration form which contains labels, textboxes, checkboxes, radi buttons, submit and reset buttons.	12	L3	CO2
	b.	Explain the following with example (i) Heading tag (ii) Hypertext Link (iii) Image tag (iv) List tag.	8	L3	CO2
Module – 2					
Q.3	a.	What is CSS? Explain the different levels of CSS with example.	10	L2	CO2
	b.	Explain any 5 array methods used in Javascript.	10	L2	CO2
OR					
Q.4	a.	List and explain different selector in CSS with suitable example.	10	L1,2	CO2
	b.	What are the different ways of accessing XHTML document elements in Javascript? Explain any two.	10	L1	CO2
Module – 3					
Q.5	a.	What is Bootstrap? What are the advantages of Bootstrap? Give an example for basic HTML template by using Bootstrap.	10	L2	CO4
	b.	Explain the Grid system in Bootstrap with example.	10	L2	CO4
OR					
Q.6	a.	Explain various types of forms in Bootstrap. Explain their use with code snippet.	10	L2	CO4
	b.	Explain with an example, the Bootstrap “Button classes” for different styles of Button in Bootstrap CSS.	10	L2	CO4

Module – 4					
Q.7	a.	What is the need of JQuery? Give the standard syntax of JQuery and explain with an example.	10	L2	CO5
	b.	Develop programs to implement the following JQuery effects : (i) hide() and show() (ii) fadeIn() and fadeOut()	10	L2	CO5
OR					
Q.8	a.	What are the Events? Explain any 5 JQuery methods to handle events.	10	L2	CO5
	b.	Write a short note on : (i) JQuery HTML (ii) JQuery CSS.	10	L2	CO5
Module – 5					
Q.9	a.	Define AngularJS. What are the features of AngularJS? Give the advantages of using AngularJS.	10	L2	CO5
	b.	Explain AngularJS controller with example.	10	L2	CO5
OR					
Q.10	a.	What are AngularJS directives? Explain varieties of directives used in AngularJS with example.	10	L2	CO5
	b.	What are AngularJS filters? Write a AngularJS program to show how you can add filters to expressions.	10	L3	CO5

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22MCA252					
OR					
Q.8	a.	Explain Bayesian classification in detail.	10	L2	CO4
	b.	Write in detail about CART (Classification and Regression Tree).	10	L2	CO4
Module – 5					
Q.9	a.	Explain data mining for business application with suitable examples.	10	L2	CO5
	b.	Write in detail about Big data business analytics.	10	L2	CO5
OR					
Q.10	a.	With suitable example, explain click stream mining.	10	L2	CO5
	b.	What is the role of data mining in CRM applications? Discuss.	10	L2	CO5

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OR

Q.6	a.	Explain the inference modes-backward chaining and forward chaining with example.	10	L2	CO2
	b.	Illustrate the following: i) Frames and frame based system ii) Bayesian network.	10	L2	CO2
Module – 4					
Q.7	a.	Discuss STRIPS with blocks world problem.	10	L2	CO2
	b.	Discuss the components of planning system.	10	L2	CO2
OR					
Q.8	a.	Discuss Winston's learning program.	10	L2	CO4
	b.	Illustrate the following: i) Machine learning ii) Adaptive learning.	10	L2	CO2
Module – 5					
Q.9	a.	Explain architecture of expert systems and role of expert systems.	10	L2	CO3
	b.	Explain meta knowledge and knowledge acquisition.	10	L2	CO2
OR					
Q.10	a.	Explain DART expert system in Artificial Intelligence.	10	L2	CO4
	b.	Discuss the use of MYCIN in medical diagnosis.	10	L2	CO4

Module – 4

Q.7	a.	What is web scrapping? Explain the python libraries used for web scrapping.	10	L3	CO4
	b.	Explain the steps involved in fetching the data by submitting the form.	10	L3	CO4

OR

Q.8	a.	Explain NumPy array attributes with example.	10	L3	CO4
	b.	Write a note on aggregation functions available in NumPy with example.	10	L3	CO4

Module – 5

Q.9	a.	Implement a python program to demonstrate data visualization using matplotlib.	10	L3	CO4
	b.	Write short notes on: i) Matplot library ii) Seaborn library.	10	L3	CO4

OR

Q.10	a.	Create a python program to demonstrate data visualization to plot a line plot, scatter plot using seaborn.	10	L3	CO4
	b.	Write a note on multiple subplots with examples.	10	L3	CO4

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Third Semester MCA Degree Examination, Dec.2024/Jan.2025
Internet of Things

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

Module – 1			M	L	C
Q.1	a.	What is IoT? Explain the evolutionary phases of the Internet.	6	L2	CO1
	b.	Discuss any two applications of IoT.	6	L3	CO1
	c.	With a neat diagram discuss the IoT World Forum (IoT WF) Standardized Architecture.	8	L2	CO1
OR					
Q.2	a.	With a neat diagram discuss the one M2M IoT architecture.	7	L2	CO1
	b.	With a neat diagram discuss a simplified IoT architecture.	8	L2	CO1
	c.	Discuss the most significant challenges and problems that IoT is currently facing.	5	L3	CO1
Module – 2					
Q.3	a.	Define sensors. Discuss different categories and types of sensors.	9	L2	CO2
	b.	Define smart objects. Discuss the characteristics of a smart object with diagram.	5	L2	CO2
	c.	What is LoRaWAN? Discuss about ZigBeeIp protocol stack.	6	L3	CO2
OR					
Q.4	a.	Define Actuators. Discuss the comparison of Sensor and Actuators functionality with humans.	6	L2	CO2
	b.	Discuss 802.15.14 physical layer and MAC layer properties along with its frame format.	8	L2	CO2
	c.	List different characteristics and attributes considered when connecting smart objects. Discuss any two.	6	L1	CO2
Module – 3					
Q.5	a.	What are the benefits of Internet Protocol?	6	L2	CO3
	b.	How optimizing IP for IoT is done? Discuss 6LOWPAN header stacks.	10	L2	CO3
	c.	What is the use of IoT Data Broker?	4	L1	CO3

1 of 2

OR

Q.6	a.	Define SCADA. Discuss the protocol stack for transporting serial DNP3 SCADA over IP.	10	L2	CO3
	b.	Why CoAP and MQTT protocols designed? Compare the differences between CoAP and MQTT.	10	L2	CO3

Module – 4

Q.7	a.	Discuss the following: i) Structured versus unstructured data. ii) Data in Motion versus data at rest. iii) “Three Vs” to categorize big data. iv) Types of data analysis results.	10	L1	CO4
	b.	Discuss massive parallel processing databases.	6	L2	CO4
	c.	Discuss about distributed Hadoop cluster.	4	L2	CO4

OR

Q.8	a.	What are the key values of edge streaming analytics? Illustrate the stages of data processing in an edge APU.	10	L1	CO4
	b.	Discuss OCTAVE Allegro steps and phases.	10	L2	CO4

Module – 5

Q.9	a.	With a neat diagram, explain Raspberry Pi board.	10	L2	CO5
	b.	Write a short note on smart traffic control and connected environment.	10	L2	CO5

OR

Q.10	a.	Discuss an IoT strategy for smarter cities.	5	L2	CO5
	b.	With a neat diagram, discuss smart city IoT architecture.	10	L2	CO5
	c.	Write a short note on Arduino.	5	L2	CO5

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22MCA332

Third Semester MCA Degree Examination, Dec.2024/Jan.2025 Cloud Computing

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is the innovative aspect of cloud computing?	10	L2	CO1
	b.	Write short notes on: (i) Web 2.0 Applications (ii) Key features of service orientation	10	L2	CO1
OR					
Q.2	a.	Write about the primary distributed computing technologies in cloud computing.	10	L2	CO1
	b.	What is utility computing? Explain in detail.	05	L1	CO1
	c.	Explain about the virtualization and its key aspect.	05	L2	CO1
Module – 2					
Q.3	a.	What is parallel and distributed computing and distinction between them?	10	L2	CO2
	b.	Write detail notes on: (i) The major categories of parallel computing system (ii) What is service-oriented computing?	10	L2	CO2
OR					
Q.4	a.	Explain about the parallel processing and its appealing computing alternatives.	10	L2	CO2
	b.	Discuss about the technologies that enable service computing.	10	L2	CO2
Module – 3					
Q.5	a.	What are the characteristics of virtual environments?	10	L2	CO2
	b.	Discuss about the classification of taxonomy of virtualization at various levels.	10	L2	CO2
OR					
Q.6	a.	Discuss about the Machine Reference Model for execution in virtualization.	05	L2	CO2
	b.	What is Xen? Discuss the elements of virtualization.	05	L2	CO2
	c.	Discuss Hyper-V's architecture. Illustrate its application in cloud computing.	10	L2	CO2
Module – 4					
Q.7	a.	What is "XaaS" abbreviation stand for? Explain in detail.	10	L2	CO3
	b.	Explain briefly about cloud reference model and essential components.	10	L2	CO3
OR					
Q.8	a.	Explain about the different forms of cloud and its classification.	10	L2	CO3
	b.	Discuss about the challenges of cloud computing. List out the application development concept.	10	L2	CO3
Module – 5					
Q.9	a.	What is AWS? What are types of services does it provide?	10	L3	CO4
	b.	Discuss the storage services provided by windows AZURE in detail.	10	L3	CO4
OR					
Q.10	a.	Describe how cloud computing technologies can be used to facilitate remote ECG monitoring.	10	L3	CO4
	b.	What are the most significant benefits of cloud technologies for social networking applications?	10	L3	CO4

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22MCA342

Third Semester MCA Degree Examination, Dec.2024/Jan.2025 Introduction to Dot-Net Framework for application Development

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	List and explain the different data types in C#.	10	L2	CO1
	b.	Write C# program using try, catch, finally to explain any predefined exception.	10	L3	CO1
OR					
Q.2	a.	Discuss in detail three different types of looping statement.	10	L2	CO1
	b.	Describe method overloading with a suitable program.	10	L2	CO1
Module – 2					
Q.3	a.	Discuss between value types of reference type and write a program to explain boxing and unboxing.	10	L2	CO1
	b.	Describe static data and static method with a suitable program. Mention its restrictions.	10	L2	CO1
OR					
Q.4	a.	What is Array? Discuss Array class with suitable example.	10	L1	CO1
	b.	Explain the usage of for. . . Each loop with an example in C#.	10	L2	CO1
Module – 3					
Q.5	a.	Describe Abstract class and Abstract method with a suitable example.	10	L2	CO1
	b.	Explain with a suitable program multiple and multiple inheritance.	10	L2	CO1
OR					
Q.6	a.	Describe the concept of interface with multiple inheritances like behavior.	10	L2	CO1
	b.	Write a program to illustrate Hybrid inheritance.	10	L3	CO1
Module – 4					
Q.7	a.	Discuss briefly about the list and Linked list class.	10	L2	CO2
	b.	Explain the different access modifiers available in C# and discuss how they control and variability and accessibility of fields within a class hierarchy.	10	L2	CO2

OR					
Q.8	a.	Demonstrate the declaration and usage of generic classes, methods and interface in C#.	10	L2	CO2
	b.	Describe the concept of indexers in C# and their role in providing array line access to objects.	10	L2	CO2
Module – 5					
Q.9	a.	Demonstrate how to implement and use iterator to iterate over different types of collection.	10	L2	CO3
	b.	Explain four steps involved in creating and using delegate in the program.	10	L2	CO3
OR					
Q.10	a.	Describe three different types of queries that can be performed using LINQ.	10	L2	CO3
	b.	Explain briefly about operator overloading with a suitable example.	10	L2	CO3

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